TSFM-Lens: Controlled, Causal and Confirmatory Interpretability for Time-Series Foundation Models

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Abstract

Time-series foundation models (TSFMs) are deployed zero-shot, yet the tools that made language models inspectable do not transfer directly: tokens span different lengths of time across architectures, no ground truth defines a forecasting "concept", and the causal effects of internal features are small enough that uncontrolled measurements mislead. We present TSFM-Lens, an open-source system that turns a list of checkpoints into one self-contained interpretability report with one command. It is controlled: a leakage-audited synthetic benchmark with known generative components, and a forecaster with planted concepts, validate every probe. It is causal: concepts are defined by what removing them does to the forecast, against random-direction nulls matched to each feature's own activation profile. It is confirmatory: exploratory claims are registered and tested once on a sealed split. Applied to seven TSFMs (TimesFM, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5, Timer and Time-MoE), it finds that causal features act mainly on forecast level and seasonality; that different architectures select the same inputs for a concept (20 of 20 registered claims, from two source concepts, confirmed on 971 held-out series and on four independent private draws); that the basic effect types are universal, with causal features that raise the level, lower it or damp seasonality found in every one of the seven models; that fine-grained concepts are nonetheless organized by model, with no strict concept spanning more than three models (confirmed held-out); that activation prominence is a poor guide to causal importance (83.5% of prominent features have no causal effect on held-out data); and that depth-resolved reaction profiles replicate for six of nine corruptions. A second, three times larger corpus confirms the same pattern (41 of 63 claims, transfer 20 of 20 again and 20 of 20 on raw real GIFT-Eval windows), and shows that neither more data, a more powerful per-feature test nor gradient-selected candidates make single features more specifically causal. Every claim carries its evidence class, and a public ledger gives each result with its reproduction command.

Keywords: mechanistic interpretability, time-series foundation models, sparse autoencoders, causal intervention, pre-registration

1 Introduction

Pretrained forecasting models are now default components. TimesFM (Das et al., 2024), the Chronos family (Ansari et al., 2024, 2025), Sundial (Liu et al., 2025), Timer (Liu et al., 2024), Moirai (Woo et al., 2024), MOMENT (Goswami et al., 2024), Lag-Llama (Rasul et al., 2023) and Time-MoE (Shi et al., 2025) are released as open checkpoints and used zero-shot. Their architectures differ sharply (decoder-only stacks over 32-step patches, encoder–decoders over quantized single steps, encoders with cross-series attention, sampled flow-matching heads), and leaderboards say which is more accurate. They do not say what these models

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represent, whether architectures this different represent the same things, or whether any internal structure we can name drives the forecast.

Mechanistic interpretability offers probes (Alain and Bengio, 2017; Hewitt and Liang, 2019), representational similarity (Kornblith et al., 2019), activation patching (Vig et al., 2020; Meng et al., 2022; Zhang and Nanda, 2024), lenses (Belrose et al., 2023) and sparse dictionaries (Bricken et al., 2023; Gao et al., 2025), packaged in libraries such as TransformerLens (Nanda and Bloom, 2022) and pyvene (Wu et al., 2024), and early work applies pieces of this to TSFMs (Wiliński et al., 2025; Mishra, 2026; Oublal et al., 2026; Divo et al., 2026). Three properties of forecasting models break the direct transfer, and each causes silent errors rather than crashes:

(i) **Tokens are not comparable across models.** One token is 1 time step in Chronos-T5, 32 in TimesFM, 16 in Chronos-2 and 96 in Timer, so every cross-model comparison needs a verified mapping from tokens to time.

(ii) **There is no ground truth for concepts.** Real series carry no labels for trend, period, changepoints or noise process, so a probe that "finds seasonality" cannot be checked.

(iii) **Effects are small and easy to fake.** Removing one sparse feature typically moves a forecast by a few hundredths of the context's standard deviation, less than the dictionary's own reconstruction error. At that scale a precision mismatch, a patch the head never reads or a null measured on the wrong rows yields a clean, confident and wrong result.

We present TSFM-Lens, a system designed around these problems, and use it to study three questions: **RQ1**, what forecast-relevant concepts TSFMs learn; **RQ2**, whether these concepts are shared across architectures, in the sense of responding to the same inputs, having the same effect, or both; and **RQ3**, whether they causally affect the forecast and how much of it they account for. A supporting question, **RQ4**, asks where in depth these computations happen and how comparable depth is across architectures.

TSFM-Lens rests on three commitments. It is *controlled*: a synthetic benchmark with exact generative ground truth and a battery of structured perturbations gives every probe a known answer. It is *causal*: a feature counts only if removing it changes the forecast more than removing a random direction of the same size on the same series, with reach controls proving the intervention landed. It is *confirmatory*: dev findings are frozen in a hash-pinned registry and tested once on a sealed private split. Measured *time alignment* makes architectures comparable, and a *one-command report* explains every figure in plain language.

Contributions.

C1. An open-source system that turns a configuration listing checkpoints into a self-contained HTML report through 19 fingerprinted stages, with a zero-code adapter path for new checkpoints (Section 4).

C2. A controlled benchmark generator with two leakage tiers, a leakage audit, sealed splits with fresh epochs and bit-exact regeneration (Section 4.1).

C3. A causal concept pipeline that separates two notions of sharing that are usually conflated: *same inputs* and *same effect on the same inputs*, validated on a forecaster with planted concepts and decoys (Sections 5.3 and 5.5).

C4. A statistical protocol with series-level resampling, a multiplicity ledger and one-shot confirmation (Section 6).

C5. A seven-model study answering RQ1–RQ4 with typed evidence and held-out replication on two independent corpora (Section 7).

Main findings. On TimesFM 2.5, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5-Base, Timer and Time-MoE:

- **RQ1: TSFMs learn a compact vocabulary of causal effects.** 442 feature-channel effects across 55 measured layers clear a profile-matched null (2.47 times the empirical chance rate, up to 5.70 times in Chronos-Bolt), and they act mostly on forecast *level* and *seasonality*, with smaller groups for trend and volatility. They form a graded continuum rather than discrete units.
- **RQ2: Architectures share inputs and basic effect types; fine-grained concepts are model-specific.** Different models select the same series for a concept: 1,296 of 1,584 dev transfer tests are reciprocal, and 20 of 20 registered transfer claims, from two source concepts, confirm on 971 sealed series and on four independent private draws, reaching all seven models. Causal features that raise the level, lower it or damp seasonality occur in every one of the seven models. At the fine grain of the strict atlas, concepts are purer by model than chance and none spans more than three models, a structural claim that confirms held-out.
- **RQ3: Causal importance is concentrated and not visible from activations.** Ten single-feature causal claims confirm on private data, each at the attainable p floor (11 on the second corpus), while 83.5% of prominent features have no causal effect; prominence does not identify what matters, and the battery does.
- **RQ4: Depth signatures are reproducible.** The agreement between TimesFM's and Chronos-2's per-corruption depth profiles replicates on held-out data for six of nine corruptions; absolute depth locations are comparable once read against captured compute (14% to 99.8% of forecast FLOPs).

Internal features also predict forecast failure beyond a strong output-only baseline for the two weakest models, a gain that confirms held-out (Section 7.6). A second run on a three times larger corpus, with more real sources and claims registered against a stricter null, reproduces the held-out pattern: 41 of 63 claims confirm, and the 20 transfer claims confirm on the new private split and on 800 raw real GIFT-Eval windows. The same run shows what limits the causal evidence: more dictionary training data, a more powerful per-feature test and gradient-selected candidates all fail to make single features more specific than on-manifold random directions (Sections 7.8 and 7.9).

Appendix A answers the questions a careful reader is likely to ask (is the data good and sufficient, is the ablation fair and statistically supported, how are token time frames aligned) with the evidence for each. The full ledger of results, each with exact numbers, evidence class, status and a reproduction pointer, is `FINDINGS.md` (indexed in Appendix B). The example

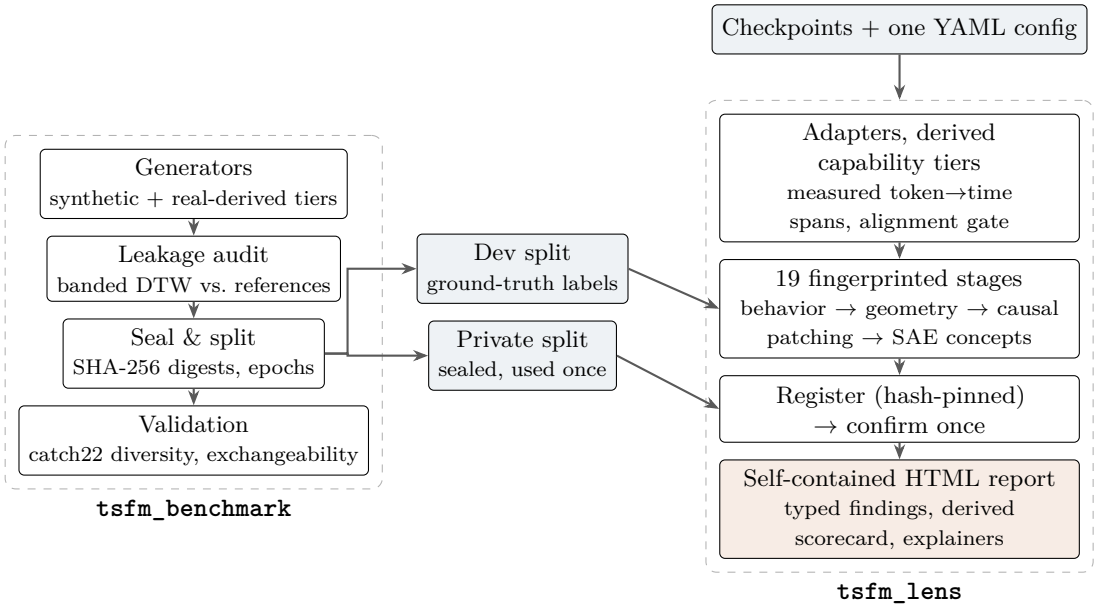


Figure 1: System overview. `tsfm_benchmark` produces leakage-audited, sealed corpora with exact ground truth; `tsfm_lens` turns a configuration naming checkpoints into one report. Dev findings are exploratory; the private split is opened once, by `confirm`, for claims registered in advance.

run of Section 7, with its report and every per-stage artifact, is `examples/panel7_v2`; `paper/figures/make_figures.py` regenerates every data figure from it.

2 Related Work

TSFMs. TimesFM (Das et al., 2024) is decoder-only over 32-step patches; Chronos (Ansari et al., 2024) quantizes values and trains a T5 encoder–decoder (Raffel et al., 2020) with a sampled decoder; Chronos-Bolt uses a direct patch head, and Chronos-2 (Ansari et al., 2025) is an encoder with cross-series group attention; Sundial (Liu et al., 2025) samples from a flow-matching head (Lipman et al., 2023); Timer (Liu et al., 2024) uses 96-step tokens, Time-MoE (Shi et al., 2025) a sparse mixture of experts over per-step tokens, and Lag-Llama (Rasul et al., 2023) reads lagged values. Most normalize each series internally (Kim et al., 2022), a detail an adapter must reproduce when it calls a model below its public API. Wan et al. (2026) document flat "collapsed" forecasts on weakly predictable targets, which we also measure (PM-07).

Interpreting time-series models. Saliency methods for time-series classifiers are unreliable across architectures (Ismail et al., 2020); Kalnāre et al. (2025) apply patching and SAEs to time-series classifiers. For TSFMs, Wiliński et al. (2025) study representational redundancy and concept steering, and four recent studies train SAEs on one TSFM (Mishra, 2026; Oublal et al., 2026; Divo et al., 2026; Yıldırım, 2026). Each examines one model or one method. None compares architectures on a common time axis, validates probes against

Capability	*TransformerLens*	*nnsight / pyvene*	*SAELens*	*Captum*	*TSFM SAE studies*[†]	**TSFM-Lens**
Hooks, caching, activation patching	✓	✓	○	○	○	✓
Sparse dictionaries and feature ablation	○	○	✓	✗	✓	✓
Time-series models without per-model engineering	✗	○	✗	○	✗	✓
Measured token→time alignment across architectures	✗	✗	✗	✗	✗	✓
Benchmark with exact generative ground truth, leakage audit	✗	✗	✗	✗	○	✓
Random-direction nulls and reach controls for every ablation	✗	✗	✗	✗	✗	✓
Cross-model concept matching (inputs vs. effects)	✗	✗	✗	✗	✗	✓
Series-level resampling and multiplicity ledger	✗	✗	✗	✗	✗	✓
Registered, one-shot confirmation on sealed data	✗	✗	✗	✗	✗	✓
Typed evidence class and generated caveat per claim	✗	✗	✗	✗	✗	✓
One-command report designed for non-experts	✗	✗	○	✗	✗	✓

Table 1: Capabilities for comparative TSFM interpretability. ✓ provided; ○ possible with user code or partial; ✗ not provided. General libraries are single-model microscopes and complementary to TSFM-Lens, which uses the same primitives. [†]Mishra (2026); Oublal et al. (2026); Divo et al. (2026); Yıldırım (2026), judged from the papers.

generative ground truth, separates input agreement from effect agreement, or tests findings on held-out data.

Methods, tooling and validity. Probes need control tasks (Hewitt and Liang, 2019; Belinkov, 2022); patching is sensitive to metrics and corruptions (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024); attention weights are not explanations (Jain and Wallace, 2019); SAE features vary across seeds (Paulo and Belrose, 2025). Representation similarity (Raghu et al., 2017; Kornblith et al., 2019; Bansal et al., 2021) underlies claims of convergence (Li et al., 2016; Huh et al., 2024). Libraries such as TransformerLens, nnsight (Fiotto-Kaufman et al., 2025), pyvene, SAELens (Bloom et al., 2024) and Captum (Kokhlikyan et al., 2020) provide primitives for one model, not comparative designs, nulls, multiplicity control or confirmation (Table 1). From empirical science we import pre-registration (Nosek et al., 2018), protection against adaptive reuse of holdouts (Dwork et al., 2015), leakage auditing (Kapoor and Narayanan, 2023) and series-level resampling (Efron and Tibshirani, 1993; Cameron et al., 2008).

3 Design Principles

Each principle is enforced in code as a gate, refusal or test, so that a measurement that cannot be trusted is withheld with a stated reason rather than reported.

P1. Ground truth by construction. Every probe is first validated on a case whose answer is known: the benchmark records every generative component of every synthetic series, and every perturbation is applied to series whose components are known.

P2. Alignment is measured, not declared. Token spans are measured from impulse responses; cross-model analysis happens only on a common window axis; models that are not time-localized receive behavioral analysis only.

P3. Causal claims need a null, a reach control and a same-seed baseline. Every intervention is compared with a random intervention of matched size on the same series; self-patching must change the forecast by exactly 0.0 and cross-patching by a nonzero amount; sampled models are seeded identically in compared calls.

P4. The series is the unit of inference. All intervals and p-values resample series; p-values are floored at $1/n_{\text{boot}}$, and each correction family's attainable minimum is checked.

P5. Explore, then confirm once. Dev results are exploratory; registered claims are tested once on a sealed split, and a second look is recorded.

P6. Every claim is typed; failures are loud. Each finding carries an evidence class (BEHAVIORAL, DESCRIPTIVE, GEOMETRIC, LINEAR-TRANSLATABLE, CAUSAL WITHIN-MODEL, ILLUSTRATIVE, HELD-OUT CONFIRMED) and a generated caveat; unsupported analyses are skipped with a rendered reason.

P7. Written for newcomers. The deliverable is one HTML file, designed with the intention that a reader new to interpretability can follow it, with every figure explained and every verdict derived from artifacts rather than written by hand.

New behavior is opt-in so sealed corpora regenerate bit-exactly, and stage configurations are fingerprinted so stale cached artifacts are refused.

4 System Architecture

The repository holds two separately installable packages: `tsfm_benchmark` (one subpackage for generation, audit and sealing, another for diversity and exchangeability) and `tsfm_lens` (subpackages for models, extraction, analysis, SAEs and the report, plus the stage orchestrator `pipeline.py`). The folder `examples/panel7_v2` contains the report, configuration and every lightweight artifact of the run analyzed in Section 7, so the report and this paper's figures can be regenerated without GPUs. The `main` branch is the clean publishable surface; a `dev` branch keeps the research log, standalone study drivers and experiment configurations.

4.1 Controlled benchmark generation

Real data enter the benchmark only as leakage references, as realism calibration and as raw material for explicitly labeled generators. The *synthetic* tier touches no real data and has exact ground truth: `parametric` is additive (trend, multiple seasonalities, AR noise, changepoints, anomalies, random walk, intermittency, heteroskedasticity), and `random_parametric` draws a recipe from one of eight frozen archetypes (four more are opt-in). The *real-derived* tier (`mixture`, `block_bootstrap` (Künsch, 1989), `sequential_par`) inherits its source dis-

tribution and has attributional ground truth. Because "no leakage" and "realistic" pull in opposite directions, each sample carries its tier label and analyses report per tier.

Every sample passes a two-stage leakage gate against the Monash archive (Godahewa et al., 2021) (a cheap prefilter, then banded DTW; Sakoe and Chiba, 1978) and a cross-split near-duplicate check. Dev and private splits use disjoint seed ranges mixed with SHA-256; every sample is hashed so that `load_sealed(verify=True)` detects tampering, and fresh private *epochs* are minted by striding the seed space. A separate validation package measures redundancy by shape matching and diversity in the catch22 feature space (Lubba et al., 2019) (participation-ratio dimension, Gao et al., 2017; nearest-neighbor tails), never on a 2-D embedding, and tests dev/private exchangeability with an energy distance (Székely and Rizzo, 2013) plus TOST equivalence (Schuirmann, 1987) (v1 benchmark: energy $p = 0.51$, 21 of 22 features equivalent, MP-09; the case study's v2 dev split against its private epoch: energy $p = 0.951$, 20 of 22 features equivalent, 6 near-duplicate pairs among 937,015).

Two perturbation families are built in: nine structured *corruptions* (noise, detrend, deseasonalize, frequency shift, level shift, spike, smooth, warp, dropout), which drive L3, and *generator counterfactuals*, which move one generative knob along a dose ladder with every random draw fixed. Dose 1.0 must reproduce the original bit-exactly, a free identity test that caught a real bug (BM-05).

4.2 Model adapters and capability tiers

A model enters through an adapter (`tsfm_lens/tsfm_lens/models/base.py`) exposing `load`, `predict` (point forecast and quantiles in the original scale), the module, input preparation, a forward pass and `token_time_spans`. The *capability tier* is derived from which methods a subclass overrides, never declared: 0 (black box), 1 (observable hidden states with spans), 2 (steerable: single-pass context, so patching is possible) and 3 (decomposable attention). Stages above a model's tier are skipped with the missing method named in the report. A missing capability (`CapabilityUnavailable`) is a fact about the adapter; a measured failure of time localization (`NotTimeLocalized`) is a fact about the model and routes it to behavioral analysis only.

New models use the zero-code `generic_hf` adapter, which discovers layers and measures spans and refuses rather than guesses (its span discovery reproduces all six hand-written adapters' spans exactly, mean IoU 1.0, and recovers Timer's 96-step spans from measurement alone; MP-01), or a copy of `tsfm_lens/tsfm_lens/models/TEMPLATE_adapter.py` in `models/contrib/`. `run.py --check-adapter` then runs patch-identity and reach conformance, the alignment gate, span discovery and capability checks. Table 2 lists the models.

4.3 Time alignment across architectures

Each token's span is measured from its impulse response (`extraction/span_discovery.py`), gated on peak-to-pedestal contrast and contiguity. Given spans I_j and windows $W_w = [wL, (w+1)L)$ ($L = 32$ by default), tokens are overlap-pooled,

$$P_{wj} = \frac{|I_j \cap W_w|}{\sum_{j'} |I_{j'} \cap W_w|}, \qquad \tilde{h}_w = \sum_j P_{wj}\, h_j, \tag{1}$$

Model	Checkpoint	Architecture	Token	Head	Tier	Params	FLOPs obs.
TimesFM 2.5	`google/timesfm-2.5-200m`	decoder-only	32	deterministic	3	231M	42.5%
Chronos-2	`amazon/chronos-2`	encoder (+group attn.)	16	deterministic	3	119M	97.5%
Sundial	`thuml/sundial-base-128m`	decoder + flow head	16	sampled	2	128M	21.3%
Chronos-Bolt	`amazon/chronos-bolt-small`	encoder + patch head	16	deterministic	2	48M	79.1%
Chronos-T5-Base	`amazon/chronos-t5-base`	encoder–decoder	1	sampled	3	201M	13.9%[a]
Timer	`thuml/timer-base-84m`	decoder-only	96	deterministic	2[b]	84M	99.8%
Time-MoE	`Maple728/TimeMoE-50M`	decoder-only, MoE	1	deterministic	2[b]	113M	50.0%
Lag-Llama	`Lag-Llama`	decoder over lags	lags	sampled	2	–	–

Table 2: Models with adapters; the first seven form the case study (Chronos-T5 also comes in other sizes, 8M–709M). "Token" is time steps per token; "FLOPs obs." is the share of a forecast's FLOPs spent in captured blocks at context 480, measured with PyTorch's FLOP counter. [a]Encoder only (DE-05). [b]Via `generic_hf`.

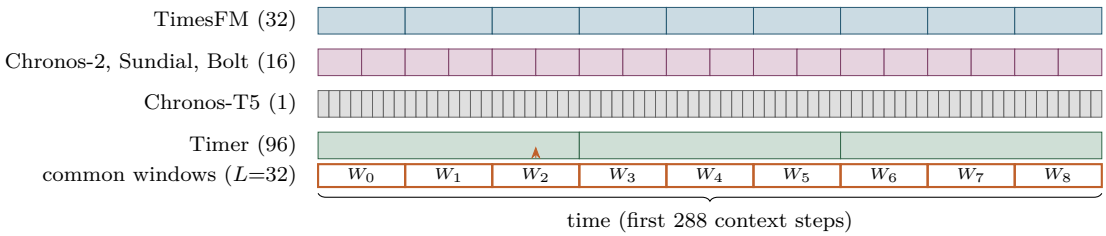


Figure 2: Time alignment. Tokens cover measured spans of different widths (Chronos-T5's one-step tokens drawn in groups of four); Equation (1) pools them onto common windows.

and every cross-model analysis runs on [series, window, dim] tensors (Figure 2); an empty window raises rather than yielding a zero row. During extraction an *alignment gate* injects an impulse into each window and checks, per layer, that the matching pooled window moves most, normalizing hits by their attainable ceiling (a 96-step token on 32-step windows reaches at most one third; MP-02), with the impulse amplitude self-calibrated per checkpoint and context length.

4.4 The stage pipeline and its gates

An analysis is a pipeline of 19 stages (Figure 3) writing typed artifacts. Stages skip themselves when their artifacts exist; each declares the configuration keys it depends on, which are fingerprinted so that a changed setting marks exactly the dependent artifacts stale. Three gates write their decisions to JSON and the report names them as skip reasons: the *tier* gate, the *run-shape* gate (comparison stages are dropped for one model) and the *routing* gate. Extraction captures the residual stream into a float16 zarr store, and every intervention goes through one primitive (`hooks.token_patch`), which raises if the model silently splits a batch.

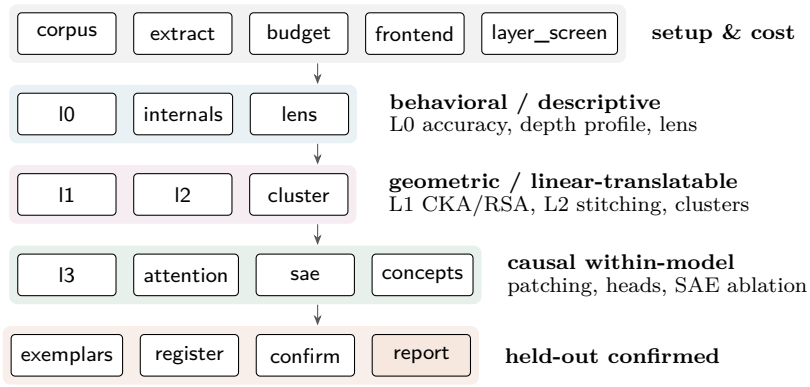


Figure 3: The 19-stage pipeline, grouped by the strongest evidence class each group can produce. Gates (tier, run shape, routing, configuration fingerprints) skip a stage per model with a rendered reason; only `confirm`, on the sealed private split, produces HELD-OUT CONFIRMED evidence.

4.5 The report: designed to be readable by newcomers

The deliverable is one self-contained HTML file (`examples/panel7_v2/report.html`), designed with the intention that a reader without interpretability training can follow it, while an expert can audit every number (Figure 4).

- **Question-first.** Seven parts pose plain questions ("Is the comparison fair?", "Where does the forecast form?", "What causally drives the forecast?", ...); each stage opens with a card stating its question, method, what good and bad results look like, and what it cannot tell you.
- **Every figure explained.** A caption plus a collapsible "What does this mean?" note (purpose, how to read, limitations): 530 notes in the example. A glossary and methods appendix are generated from the code.
- **Derived verdicts.** The scorecard is a pure reduction over artifacts with no model names hard-coded; each row shows the value, its reference (null, floor, ceiling or baseline) and the rule that yields the verdict.
- **Typed findings.** Each claim is a `Finding` record with evidence class, plain-language sentence and generated caveat: 240 in the example (89 descriptive, 54 behavioral, 49 causal, 42 linear-translatable, 4 geometric, 2 illustrative).
- **Nothing silently missing.** Skipped stages render their gate reason, failed sections render visibly, and a multiplicity ledger (123 families, 4,238 comparisons) flags families that cannot reach α.

The complete analysis is one command,

```
python run.py --config configs/examples/large_4model.yaml
```

and smaller presets cover a CPU-only mock smoke test, a single model and a pair.

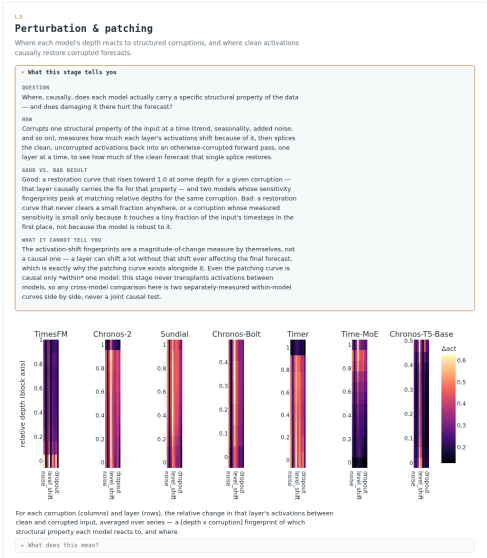


Figure 4: Excerpt of the generated report (L3 section of the seven-model example report, `examples/panel7_v2`): the stage card states the question, method, how to read good and bad results and the limitation, above the per-model corruption-sensitivity fingerprints. Every figure carries a caption and an expandable explanation.

5 Measurement Methods

Throughout, $x \in \mathbb{R}^T$ is a context ($T = 480$ in the case study), $y \in \mathbb{R}^H$ its continuation ($H = 64$), $\hat{y}$ a point forecast and $h^{(\ell)}$ the residual stream after block ℓ.

5.1 Behavior, geometry and perturbation (L0–L3)

L0. Accuracy is MASE (Hyndman and Koehler, 2006) per series, with the seasonal lag from ground truth and a scale floor on the naive denominator, without which intermittent series spuriously improve (MN-23); sMAPE, pinball loss (Gneiting and Raftery, 2007) and calibration are reported alongside. Pairs are compared by a paired series bootstrap per family, Holm-corrected across all pairs and families (Holm, 1979), and read against each model's repeat-run noise floor. `budget` measures parameters, FLOPs, latency and captured-FLOP share; `frontend` tests scale equivariance, NaN handling and quantization before any layer runs.

L1, L2. For aligned, window-pooled and centered activations A, B over the same series, linear CKA (Kornblith et al., 2019) $\|B^\top A\|_F^2 / (\|A^\top A\|_F \|B^\top B\|_F)$ is computed for every layer pair of every model pair, with a cluster bootstrap and a shuffled-series null. L2 fits ridge maps between layers and reports only the *gain* over an input-feature baseline, since much of any layer is linearly predictable from the input. Cluster agreement uses AMI (Vinh et al., 2010). These claims are GEOMETRIC or LINEAR-TRANSLATABLE.

L3. For each corruption c, the sensitivity *fingerprint* is the relative activation change $\|h^{(\ell)}(c(x)) - h^{(\ell)}(x)\| / \|h^{(\ell)}(x)\|$ per layer, and the *restoration curve* is the fraction of forecast

damage undone by patching clean activations of layer ℓ (whole, or per window) into the corrupted pass. Models are compared by rank-correlating fingerprints on shared relative depth, never by transplanting activations. Restoration is CAUSAL WITHIN-MODEL; cross-model agreement is correlational.

5.2 Depth

A skip lens decodes the forecast from each layer through the model's head and a tuned ridge lens (Belrose et al., 2023) fits per-layer readouts; `internals` reports effective dimension, family decodability and CKA to the input. The depth axis is the fraction of the *whole* stack, including uncaptured parts, so Chronos-T5's last captured block sits at 0.478 (DE-04); and the lens is withheld for a model whose head reads positions the patch never writes, where it would return a flat, meaningless curve (DE-03).

5.3 Causal concepts

Sparse dictionaries. Each target layer gets a TopK SAE (Makhzani and Frey, 2013; Gao et al., 2025), $z(h) = \mathrm{TopK}(W_e(h - b_d) + b_e)$, $\hat{h} = W_d z(h) + b_d$, with searched dictionary size, an auxiliary loss reviving dead atoms and a minimum step count (MP-03). Since single seeds are not reproducible (Paulo and Belrose, 2025), replicate dictionaries give a within-model reproducibility ceiling (MP-04).

The ablation battery. Feature i is tested on the k series where it fires most, by comparing the forecast under the reconstruction with $z_i = 0$ against the full reconstruction:

$$\Delta_i^{(c)}(x) = g_c(f(x; h^{(\ell)} \leftarrow \hat{h}_{-i})) - g_c(f(x; h^{(\ell)} \leftarrow \hat{h})), \tag{2}$$

where g_c is one of nine forecast channels (trend, seasonal strength, spectral centroid, level, dispersion, near/far-horizon shape distance, MASE, flatness). Using the reconstruction as baseline keeps the SAE's error out of both arms; this matters because that error moves the forecast several times more than a typical feature (Section 7.4). Subtracting the feature's contribution from the clean activation instead would differ only through the model's nonlinear response to the error. The null is *profile-matched*: on the *same* rows, each of 16 draws removes, at every token t, exactly the amount the feature removes, $|z_i(t)|\,\|w_i\|$, along a random decoded direction, so only the direction differs. A feature clears channel c if mean $|\Delta_i^{(c)}|$ exceeds the pooled null's 95th percentile $q_{95}^{(c)}$, and its channel vector is $v_i^{(c)} = \overline{\Delta_i^{(c)}}/q_{95}^{(c)}$. Shape channels are distances and are treated as undirected. Because a mean over rows is compared with a percentile of single rows, the chance rate is not 5%; it is estimated per cell by scoring each null draw against the others (0.3–4% per target on the case study). Matching the profile is what lets the null pass the known-answer test of Section 5.5, where a null that removes one uniform, batch-averaged amount does not (MN-29). A *reach probe* first requires self-patching to change the forecast by exactly 0.0 and cross-layer patching by a relative $> 10^{-3}$; otherwise the target is withheld with its reason (Figure 5).

From features to concepts. Causal features are grouped at three granularities. *Per-target concepts* cluster one layer's vectors with k-means, choosing k by silhouette (Rousseeuw, 1987) among values whose smallest cluster has $\geq$3 members (without it, the four highest

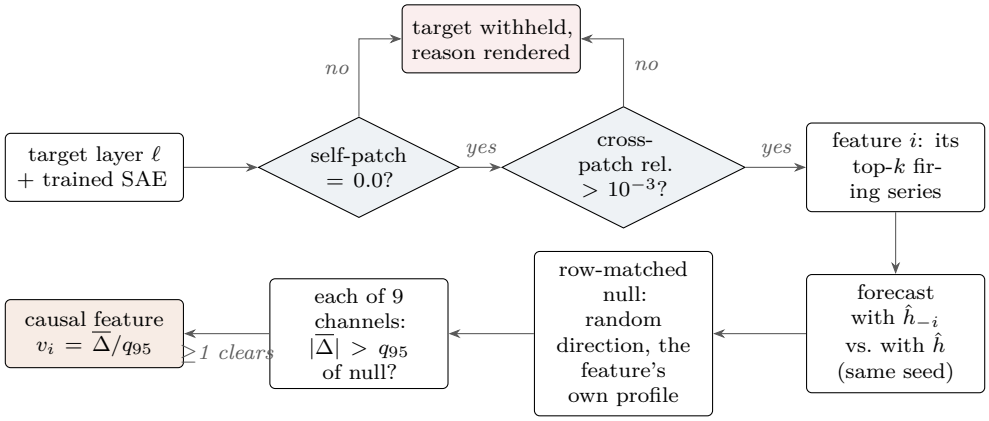


Figure 5: The ablation battery. A target is scored only after both reach controls pass; each feature is compared, channel by channel, with a random-direction null on exactly its own top-firing series.

silhouettes on the earlier four-model run, 0.607–0.717, were all one-outlier splits; SH-20); a target with no admissible k is *non-modular*. The *atlas* pools features from all models and layers and cuts a complete-linkage dendrogram at cosine 0.9, so an atlas concept is a shared causal-effect profile. *Families* are a coarser readable tiling (average linkage, cosine 0.5), tested against a column-shuffle null and titled deterministically from their channels.

5.4 Two notions of sharing

"Shared" is two questions, answered separately. *Same inputs*: concept transfer asks whether a feature in another model separates a concept's top-k series from the rest (AUC), against stratum-matched random subsets, in both directions (*reciprocal* if both clear), with Benjamini-Hochberg (Benjamini and Hochberg, 1995) per ordered model pair and leg. Input agreement between concept parts is top-k overlap (hypergeometric), never whole-series correlation: two sparse features can have $\rho = 0.80$ with no shared top series (MN-19). *Same effect on the same inputs*: every reciprocal-FDR transfer is ablated on both sides on the same series set U; each side must clear its own null on U, and level concordance and shape cosine must beat both sides' activation-matched random-feature floors. Falling below the floors' 5th percentile is "acts differently"; merely failing the 95th is "no specific agreement" (MN-18).

The report combines these into a six-rung ladder per concept (`report/derived.py`): L1 same forecast effect (atlas), L2 same inputs, L3 reproducible across SAE seeds, L4 other dictionaries select the same inputs, L5 same causal effect on the same inputs, L6 confirmed on private data. Each atlas concept also gets a `sharing_class`: *shared* (same effect and inputs), *partially shared*, *convergent* (same effect, different inputs) or *single-model*; a part is provenance-driven when $\geq$80% of its top series come from one real-derived generator.

5.5 Known-answer validation

Every causal count in Section 7 rests on a null, and a null can be wrong without any error being raised. TSFM-Lens therefore ships a forecaster whose answer is known, `mock_planted`.

One planted block writes known unit directions into the residual stream on the series that have a property (thresholded readouts of the patch mean, slope, residual deviation, seasonal projections and a spike statistic, each firing on 10–30% of the corpus), and the head applies a known effect (level, trend, seasonality or dispersion) to the forecast. It carries decoys written like concepts (*input-only*, zero causal effect; *sub-null*, weak effect), a control layer after the planted block with no planted effect, and, for the cross-model rungs, a pair of architectures (widths 64 and 96, depths 5 and 7, rotated residual basis) sharing planted concepts plus an opposite-effect twin. A scorer reads only the planted manifest and the run's artifacts, against a gate fixed before any data was seen: per-cell false-positive rate on the control layer at most 0.10 and sensitivity at least 0.5, on five construction seeds, seed 0 used to tune the attribution rule and seeds 1–4 held out. The gate is evaluated at one effect size (dose 1).

The shipped profile-matched null passes the gate on all five seeds: per-cell false-positive rate at most 0.003 and sensitivity 0.70–1.00 (a random decoded direction overlaps the planted span more than a decoy does, so the null may be conservative in the planted world). The test has teeth: a simpler null that removes one uniform, batch-averaged amount fails it on every seed (false-positive rate 0.168–0.248), because a strong or dense atom clears it on size alone. On seed 0 the profile-matched null also keeps all sub-null decoys from clearing, lifts the atlas adjusted Rand index from 0.336 to 0.469 and raises sharing-class accuracy from 0.364 to 0.714. Numbers cited from the earlier four-model run used the simpler null and are upper bounds (re-scoring that run cut its causal features from 293 to 110; MN-29). For the cross-model agreement rung of Section 5.4, the same device measures both error rates: the rung never calls a decoy "same" (0 of 9), so its positive verdicts are specific, and it detects a shared single-direction concept in 0.4 of seeds and one spread over 3–4 directions in 0.0, and quadrupling the shared series does not change this (MN-30). Absent "same" verdicts are therefore not read as differences; a redesigned subspace test also fails its pre-fixed gate (MN-32, Section 9). Known-answer tests establish what a method can detect, not that it is right on real models.

6 Statistical Protocol

Four mechanisms control how often a report with thousands of comparisons is fooled (Table 3).

The series is the resampling unit. Windows of a series share its trend, noise and scale, so every interval and p-value resamples whole series (Efron and Tibshirani, 1993; Cameron et al., 2008): paired bootstrap for accuracy, cluster bootstrap for CKA, validation-series intervals for stitching, paired cluster bootstrap for patching. Splits are by series and stratified, never head slices of a task-grouped corpus.

A multiplicity ledger. Every test belongs to a named family (Holm for confirmatory families, BH for discovery-oriented atlas transfers). Resampled p-values are floored at $1/n_{\text{boot}}$ (Phipson and Smyth, 2010), so a Holm family of m tests cannot go below m/n_{boot}; the ledger computes this minimum per family and flags families that can never be significant, a failure that grows quadratically with the number of models. The example has 123 families and 4,238 comparisons in the ledger (the 8,705 feature-channel cells of the ablation battery are judged instead by their count against the empirical chance rate, and shared-input tests

Claim type	Test statistic	Null / reference	Correction	Held-out
Model A more accurate (per family)	paired ΔMASE, series bootstrap	zero difference; repeat-run noise floor	Holm over all pairs × families	✓
Layers share geometry	linear CKA, cluster bootstrap	shuffled-series null at the same layer pair	per model pair	✓
Layer linearly translatable	ridge stitching gain	input-feature baseline (gain = 0)	per pair	–
Depth reaction agrees	Spearman ρ of fingerprints on shared depth	bootstrap CI of ρ	per corruption	✓
Layer restores forecast	restoration fraction	self-patch = 0, cross-patch ≠ 0	per model	–
Feature is causal	mean $\lvert\Delta^{(c)}\rvert$ on own rows	profile-matched random-direction q_{95}	per channel (count vs. chance); Holm when registered	✓
Concept reproducible	best-match cosine across seeds	within-model replicate ceiling	–	–
Same inputs	top-k AUC in other model	stratum-matched random subsets	BH per ordered pair and leg; Holm when registered	✓
Same effect on same inputs	level concordance; shape cosine	activation-matched feature-set floors (q_{95} agree, q_{05} differ)	per test; Holm when registered	✓
Concepts are cross-model	atlas centroid cosine and member pairs	column-shuffle and model-label nulls	Holm when registered	✓
Internals predict failure	gain in log-MASE Spearman over a free baseline	zero gain	Holm when registered	✓
Families are real	number and coverage of families	column-shuffle null	–	–

Table 3: Every claim type the case study reports, the statistic that tests it, the reference it must beat, its multiplicity correction and whether the `confirm` stage currently re-tests it on private data. Claim types without a held-out mark are reported as exploratory.

by per-test floors); its strictest L0 family (105 tests over 21 pairs) has a minimum Holm p of 0.021 at $n_{\text{boot}} = 5000$.

References from the run itself. Effects are read against repeat-run noise floors, shuffled-series nulls, input-feature baselines, random-direction nulls, activation-matched floors and replicate-seed ceilings. Before the private split is opened, `confirm` simulates each claim's minimum detectable effect (e.g. ΔMASE 0.133 at $n = 290$, 80% power), so a non-replication reads as "absent" or "underpowered".

Register, then confirm once. `register` freezes hypotheses from dev results together with the content hash of each source artifact, and hashes the registry. `confirm` refuses mismatched hashes, requires the private seal to verify, tests each claim once and refuses to overwrite its artifact; a forced rerun is recorded as `repeated_look`, and a consumed split is replaced by a fresh epoch, checked for sample-hash overlap with every earlier split of every corpus version (an epoch minted for the case study was rejected unopened because 716 of its 961 series regenerated a split an earlier confirmation had consumed). This is pre-registration (Nosek et al., 2018) in software, guarding against adaptive holdout reuse (Dwork et al., 2015); unregistered findings cannot be rescued.

Model	median MASE	80% coverage	scale resid. (s.d.)	NaN input	latency (ms)
Chronos-2	**1.2002**	0.757	0.0102	handled	23
TimesFM	1.2560	**0.779**	1.6×10^{-6}	propagates	39
Sundial	1.4131	0.360	0.0310	propagates	62
Chronos-T5-Base	1.4231	0.436	2.7×10^{-7}	handled	975
Chronos-Bolt	1.5487	0.732	0.0101	handled	15
Timer	1.9306	–	0.0262	propagates	5
Time-MoE	2.8642	–	0.0368	propagates	103
naive / seasonal naive	2.7593 / 1.9819				

Table 4: Behavior on the v2 dev corpus (965 series). Scale residual: error of $f(cx)/c$ against $f(x)$ in context standard deviations, $c = 0.001$. Latency: median time of a forecast for a batch of 8 series. Timer and Time-MoE output point forecasts only. Held-out, Chronos-2 beats TimesFM on the registered `mixture` family (paired ΔMASE 0.209 [0.119, 0.313], $p_{\text{Holm}} = 0.0002$, $n = 290$) and, with a smaller effect, overall (-0.070 [-0.129, -0.008], $p = 0.025$, $n = 971$).

7 Case Study: Seven Time-Series Foundation Models

7.1 Setup

We analyze seven models with the preset `tsfm_lens/configs/panel7_v2.yaml`: TimesFM 2.5 (Das et al., 2024), Chronos-2 (Ansari et al., 2025), Sundial (Liu et al., 2025), Chronos-Bolt and Chronos-T5-Base (Ansari et al., 2024), Timer (Liu et al., 2024) and Time-MoE (Shi et al., 2025); the last two load through the zero-code `generic_hf` adapter. The dev corpus has 965 series in five generator families; the real-derived ones draw on Monash weather, ETT (Zhou et al., 2021) and the Chronos electricity and traffic collections. Contexts are 480 steps (a multiple of Timer's 96-step token), the horizon 64, the window 32. SAEs are trained on 56 target layers (TimesFM 10, Chronos-2 9, Sundial 9, Chronos-Bolt 4, Timer 6, Time-MoE 9, Chronos-T5-Base 9) with three replicate seeds each, and $n_{\text{boot}} = 5000$. The private split is 971 sealed series (v2 epoch 3), opened once; its provenance is in Appendix D. The run recorded 21.1 hours of stage time on one RTX A5000 (24 GB). Artifacts are in `examples/panel7_v2/run`. Ledger-tagged numbers measured in another run say so (the earlier four-model run is `examples/concept_atlas_v2`); every other number is from this run or its confirmation.

Table 4 gives the models' behavior. Chronos-2 is the most accurate; Time-MoE is less accurate than the naive forecast and Timer close to the seasonal naive; Sundial's and Chronos-T5-Base's nominal 80% intervals cover 36% and 44% of outcomes; and TimesFM, Sundial, Timer and Time-MoE propagate an injected NaN silently.

7.2 RQ1: What forecast-relevant concepts do TSFMs learn?

Causal features exist, and the battery tells models apart. 194 SAE features across 55 measured targets clear the profile-matched null on at least one channel (against at most 144.95 expected by chance at the feature level, an upper bound that treats channels as independent). Fifty-five of 56 targets passed both reach controls (e.g. TimesFM `stacked_xf.10`: self-patch 0.0, cross-patch relative reach 0.253); Chronos-2's final block was withheld because cross-

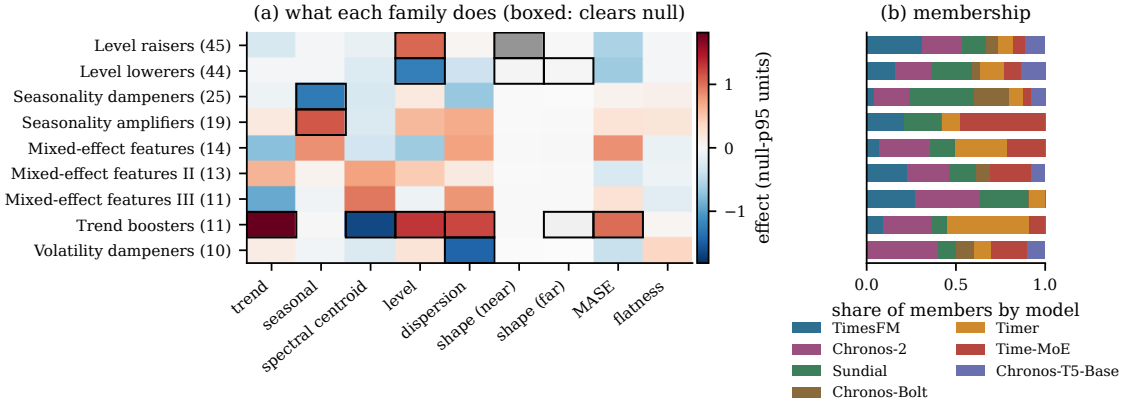


Figure 6: The nine causal-effect families. (a) Family centroid in null units, signed as what the feature *does* (red raises, blue lowers; boxed cells clear the null; shape distances unsigned, gray). (b) Membership by model. Source: `sae/concept_families.json`.

patching moved the forecast by exactly 0 (its head reads placeholder tokens a context patch never reaches; DE-03). Against the leave-one-draw-out chance rate, 442 (feature, channel) cells clear the null where 178.75 are expected (2.47 times chance), from 1.02 times (Timer) to 5.70 times (Chronos-Bolt); 22 of 55 targets are individually significant (BH at 0.05) and Timer has none, although its layers reach the forecast (relative reach 0.066–0.261; PM-17).

The concepts are about level and seasonality. Nine effect families tile the causal features (Figure 6). The two largest act on level: *Level raisers* (45) and *Level lowerers* (44), 89 of the 192 assigned features. Seasonality dampeners and amplifiers follow (25 and 19); three mixed-effect families (14, 13, 11), trend boosters (11) and volatility dampeners (10) are smaller. Level is not everything: on the earlier four-model run (original null, an upper bound) the median pure-level share of an effect was 0.535 (CA-07).

The concepts are graded, not discrete. Within a layer, 50 of 56 targets are non-modular (10 per-target concepts in total, on 5 targets, 4 of them on Timer targets whose clears are not above chance); the strict atlas assigns only 35 of 194 features to its 11 concepts, more than a column-shuffle null produces (null 95th percentile 5.05 concepts, assigned fraction 0.088 against 0.180; $p = 0.005$) but few; and the families cover 192 of 194 features but do *not* beat the shuffle null ($p = 0.68$ for their number, 0.23 for coverage). Silhouette is 0.25–0.32 for every cut from 6 to 16 families, and on the earlier run the ablation fingerprints had participation ratio 2.09 (CA-10, MN-07). Families are thus a reading vocabulary, not discovered units.

Some features align with named generative factors. After dead-feature revival, 154 of 10,240 TimesFM and 115 of 6,144 Chronos-T5-Base features match a ground-truth field beyond a permutation null (best: intermittency, $\rho = 0.881$; MP-06). The alignment is correlational: moving one generator knob does not move a matched feature's forecast effect beyond chance (MN-17), and seed-stability checks filter vivid but unstable candidates (PM-11).

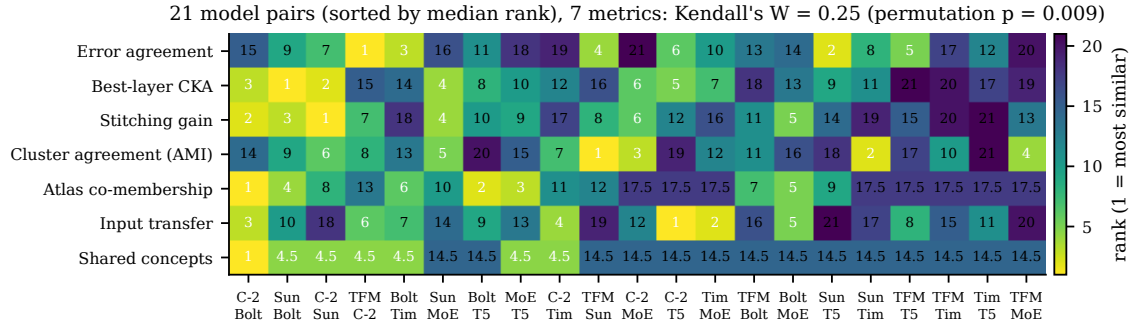


Figure 7: Rank of each model pair (columns, sorted by median rank; TFM TimesFM, C-2 Chronos-2, Sun Sundial, Bolt Chronos-Bolt, Tim Timer, MoE Time-MoE, T5 Chronos-T5-Base) under seven similarity metrics (rows); 1 is most similar. Source: `report/model_similarity.json`.

Answer to RQ1. TSFMs learn a compact vocabulary of causal effects: sparse directions that control forecast level and seasonality, and in smaller groups trend, volatility and mixtures. These effects form a continuum with a few dominant axes rather than discrete units, which is why TSFM-Lens reports families as a reading vocabulary. The battery discriminates between models, from 5.70 times chance in Chronos-Bolt to chance level in Timer on this corpus.

7.3 RQ2: Are concepts shared across architectures?

Geometry is shared above chance, but similarity has no single meaning. Peak CKA (the maximum over all layer pairs) between TimesFM and each other model is 0.342–0.426 (shuffled-series nulls 0.011–0.051); the TimesFM/Chronos-2 peak replicates held-out (dev 0.4257; private 0.4109 [0.4014, 0.4266]), and the band holds across five architectures including Timer and Lag-Llama (SH-01). Yet seven pair metrics rank the 21 pairs only weakly consistently (Kendall's $W = 0.25$, $p = 0.0095$; Figure 7): TimesFM/Chronos-2 is first by error agreement (0.948) and 15th of 21 by CKA, where Sundial/Chronos-Bolt is first (0.927), so the metrics are never pooled. Two finer-grained measures do not pass their controls on real activations (an untrained pair scores higher than a trained related pair, MN-01; a unit-correspondence measure rises under rotation, SH-19), so the report uses CKA against shuffled-series nulls and never reads it as lineage.

Models agree on which inputs a concept groups. 1,296 of 1,584 atlas transfer tests are reciprocal after BH correction. Twenty claims registered from dev alone were all confirmed on the 971 private series (Holm, maximum adjusted $p = 0.0100$, the floor for 20 claims at 2000 draws; AUC 0.919–0.970 against null 95th percentiles of 0.70–0.75; Figure 9b), replicating on four private draws (19/20, 20/20, 20/20 on earlier draws, and 20/20 here; SH-18, CA-11). This is an input-level claim. The 20 claims come from two concepts (TimesFM `xf.18`, 14 claims; Sundial `layers.6`, 6) and reach destinations in all seven models, with only 8 of 20 private best features identical to dev: the destination has *some* feature selecting the same series.

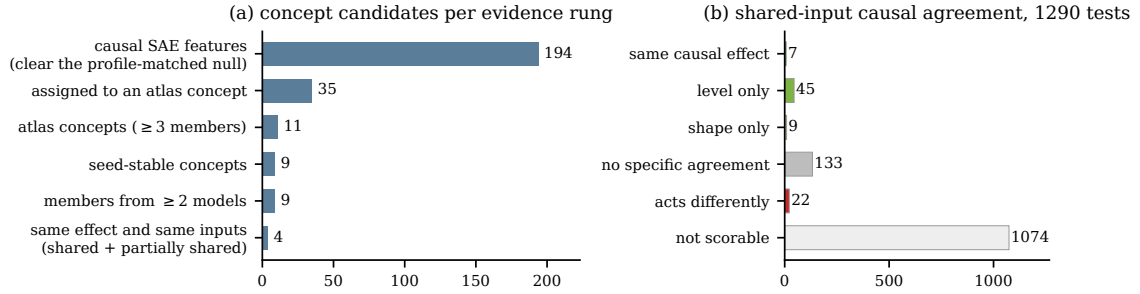


Figure 8: (a) Concept candidates at each rung (first three bars in sequence; last three are properties of the 11 atlas concepts). (b) Shared-input causal agreement over 1,290 tests drawn from the 1,296 reciprocal transfers. Source: `sae/concept_stage.json`.

The basic effect types are universal. The three largest families are found in every model: *Level raisers* (45 features), *Level lowerers* (44) and *Seasonality dampeners* (25) each have causal members in all seven models, and *Volatility dampeners* in six (Figure 6b). Membership is a property of each feature's own measured effect, so it does not depend on whether the family partition beats its shuffle null; it is descriptive and exploratory, and Timer's members should be read with its chance-level clear rate in mind.

Fine-grained concepts are organized by model. Of 11 atlas concepts, 9 have causal members from ≥2 models, but no concept spans more than 3 of the 7 models, and the clusters are *purer* by model than random assignment: 9 multi-model concepts against a null mean of 10.8 ($p_{\text{below}} = 0.025$) and mean model purity 0.591 against 0.474 (SH-22). By sharing class, 2 concepts are *shared* (same effect and inputs), 2 partially shared, 5 *convergent* (same effect reached on different series) and 2 single-model (Figure 8); among the 9 seed-stable concepts the counts are 2, 1, 4 and 2. At this grain (cosine 0.9 between null-normalized effect vectors) each architecture builds its own variants, and several arrive at the same effect independently.

Where the same effect is measured on the same series, it is detected within the Chronos family. The shared-input test ablates both sides of a reciprocal transfer on the same series and requires each to clear its own null. Of 1,290 tests (a cap of 60 per ordered pair), 216 are scorable: 7 show the same causal effect, 54 agree on level or shape alone, 133 show no specific agreement and 22 act differently. All 7 "same" verdicts are Chronos-2/Chronos-Bolt, six through one Chronos-Bolt block-4 feature set. The test is specific (it never calls a planted decoy "same"), so these detections stand; it is also insensitive across architectures (MN-30), and 1,053 of the 1,074 unscorable tests fail because a side's own ablation does not clear its null on its 21–40 shared series (median 32), so we do not read the absence of "same" elsewhere as a difference.

Answer to RQ2. Different TSFMs *select* the same series for a concept, reliably enough to replicate on four sealed private draws and in all seven models. The basic effect types (raising or lowering the level, damping seasonality) are present in every model, and the centroid effect profiles of multi-model concepts replicate held-out (Section 7.7). At the fine grain, concepts are organized by model: no strict concept is common to the panel (confirmed held-out), several converge on the same effect from different series, and the

clearest same-effect-on-same-inputs case is within the Chronos family. Whether architectures from different families share effects series by series is not measurable by ablation at these effect sizes (Section 9). Whole-layer similarity cannot reveal any of this, since its metrics disagree with one another.

7.4 RQ3: Do concepts causally affect the forecast, and how much?

Causal importance is concentrated in a minority of features, and identifying that minority requires the battery. Of the 32 single-feature causal claims registered from dev, 31 were testable on the private split and 10 confirm (TimesFM 3 of 4, Time-MoE 2 of 7, Sundial 2 of 4, Chronos-Bolt 2 of 6, Chronos-2 1 of 7, Chronos-T5-Base 0 of 4; by channel, near-horizon shape 4 of 8, MASE 3 of 8, spectral centroid 2 of 4, dispersion 1 of 3, seasonal 0 of 4, level 0 of 4); each confirmed claim has $p = 0.001$ after an adaptive redraw to 1000 null draws and $p_{\text{Holm}} = 0.031$, the floor for 31 tests. The other 21 are not detected on the new series (19 of them had not reached the 200-draw p floor on dev; one flips sign, Chronos-T5-Base block 10). Activation prominence does not identify the causal minority: 0.835 of the 927 prominent features of the private atlas are causally null (one-sided lower bound 0.817, dev 0.810, $p_{\text{Holm}} = 0.0005$; MN-14), so TSFM-Lens ranks features by measured effect rather than by activation. On the earlier four-model run the median single-feature effect was 0.0395 context standard deviations, 3.5 times below the 0.139 moved by substituting the SAE reconstruction (MN-15). Attention mass on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated ($\rho = 0.026$ TimesFM, -0.118 Chronos-T5-Base; CA-04; cf. Jain and Wallace 2019) (a minimal seasonality circuit is one head in TimesFM, five non-additive heads in Chronos-T5-Base; CA-03).

Answer to RQ3. Features and heads causally shape the forecast, and the strongest single-feature effects replicate on new series at the attainable significance floor. Contributions are small and distributed, and neither activation prominence nor attention mass locates them; a causal battery does.

7.5 RQ4: Where in depth, and how comparable is depth?

For TimesFM/Chronos-2, 6 of the 9 per-corruption fingerprint agreements replicate held-out (overall $\rho = 0.524$ [0.504, 0.544]; Figure 9a): the models react at matching depths to detrending (private 0.368), dropout (0.510) and, more strongly than on dev, frequency shift (0.381 against 0.216), and at opposite depths to noise (-0.743), spikes (-0.785), warps (-0.651) and smoothing (-0.581 against -0.418); deseasonalizing is near zero (0.086 [-0.027, 0.186]). The three non-replications are the two stronger-on-private effects (frequency shift, smoothing) and level shift, whose sign flips (dev -0.188, private 0.109 [-0.025, 0.159]); under the registered rule, dev ρ inside the private interval, a stronger private effect also counts as a non-replication. In TimesFM, per-window restoration concentrates just before the horizon and grows with depth (earlier four-model run, `noise`: 0.068 at layer 0 to 0.412 at layer 16; exploratory, $n \approx 300$; DE-01). Depth *locations* need two qualifications that every report renders: captured compute ranges from 13.9% (Chronos-T5-Base) to 99.8% (Timer) of a forecast (DE-05), and Chronos-T5's last encoder block sits at 0.478 on the

whole-stack axis (DE-04). No selected layer was robust across three layer selectors for any model (DE-08).

Answer to RQ4. Depth-resolved reaction profiles are largely reproducible (6 of 9 corruptions); absolute depth locations are comparable only after accounting for captured compute and the depth convention.

7.6 Can looking inside predict forecast failure?

If internal features are causal, do they tell a user when to distrust a forecast, or which model to use? For each model we predict a series' log MASE (Spearman) and failure flag (AUROC) from free output-only features (quantile width, flatness, catch22 of the context), then add the SAE-family activations at the last window (cross-fitted, series bootstrap, 941 dev series, $n_{\text{boot}} = 1000$; evidence class *predictive*, since a feature that predicts error is not a cause of it). The free baseline is strong (Spearman 0.76–0.85). For five of seven models (Chronos-2, TimesFM, Sundial, Chronos-T5-Base, Chronos-Bolt) the gain's CI includes zero for both tasks (log-MASE Spearman gains -0.004 to $+0.005$, AUROC -0.010 to $+0.008$). It is positive for the two weakest models, Timer ($+0.030$ [0.019, 0.043], baseline 0.787, failure rate 0.30) and Time-MoE ($+0.012$ [0.005, 0.019], baseline 0.853, failure rate 0.62), robust across five fold seeds, and no failure-AUROC gain excludes zero. Both gains were registered and confirm when the frozen procedure is refit on the private split ($n = 935$): Timer 0.0446 [0.0292, 0.0605], $p_{\text{Holm}} = 0.002$; Time-MoE 0.011 [0.0032, 0.0194], $p_{\text{Holm}} = 0.012$ (14 dev comparisons were run and 2 selected, which is legitimate because the private split referees). Routing each series to the model with the lowest predicted MASE does not help: baseline routing reaches mean MASE 1.948 [1.814, 2.070], baseline plus internals 1.964 [1.837, 2.093] (gap -0.016 [-0.044, 0.010], $p = 0.258$), and both are worse than always using Chronos-2 (1.856; $p = 0.001$; oracle 1.523). Internals add information exactly where the free baseline is weakest.

7.7 What replicated on held-out data

One look at a fresh private corpus (v2 epoch 3: 971 series, no sample shared with any other split, seal verified) tested a registry of 128 hypotheses, 82 of them replicable, each Holm family per claim type as registered (Figure 9). **Confirmed:** the accuracy claim (Chronos-2 better on `mixture`, ΔMASE 0.209 [0.119, 0.313], $p_{\text{Holm}} = 0.0002$, $n = 290$; minimum detectable effect 0.133), the peak-CKA pair, 6 of 9 perturbation agreements, all 20 transfer claims, 10 of 31 testable single-feature causal claims, both structural claims (no concept spans the panel: the private atlas has 10 concepts, at most 3 of 7 models each; most prominent features are causally null) and both failure-prediction claims. **Not confirmed:** the cross-model causal claims, namely the multi-model atlas concepts under the registered conjunctive rule (0 of 3: their private centroid cosines, 0.955–0.988, beat the null's 95th percentile of 0.854–0.907 with $p \leq 0.007$, but the rule also required half of the member pairs to reach cosine 0.9, and with three members only one pair did; the centroid component alone is a post hoc reading) and shared-input agreement (0 of 6 tested, 5 of 11 not testable; the best, Chronos-Bolt block 4 to Chronos-2 block 7 "same", reproduces its verdict with $p = 0.011$ but misses Holm at 0.066). In all, 43 claims confirm, 33 do not and 6 cannot be

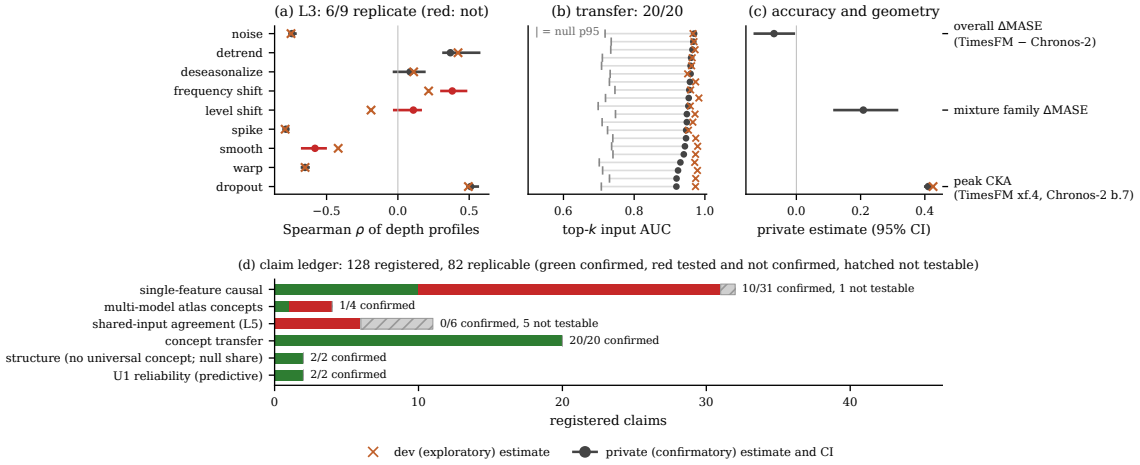


Figure 9: Claims re-tested on the sealed private split (971 series; panel a uses 800 of them). (a) Per-corruption depth-profile agreement, TimesFM vs. Chronos-2 (red: the dev value is not inside the private interval); (b) the 20 concept-transfer claims with private null 95th percentiles; (c) accuracy and peak CKA; crosses: dev, dots and bars: private with 95% series-bootstrap intervals. (d) The ledger of registered concept claims by type: confirmed, tested and not confirmed, not testable (the one confirmed atlas claim is single-model). Source: `confirm/confirmation.json`.

tested. Held out, models share what they select, the average effect of a multi-model concept reproduces, and fine-grained concepts stay model-specific; series-by-series effect agreement across families is beyond what ablation can measure at these effect sizes (Section 9). One procedural event, a crash before any statistic was written, is disclosed in Appendix D.

7.8 Replication on a three times larger corpus

To test whether these conclusions depend on the corpus, the whole pipeline was run again on a new benchmark, with every decision fixed before its data existed (preset `tsfm_lens/configs/panel7_v3.yaml`). The dev split has 2,876 series: 2,091 synthetic and 785 real-derived from 15 real sources, adding ERCOT load, M4 hourly, solar, taxi, KDD Cup air quality and pedestrian counts to the v2 sources. SAEs are trained on 50 targets, and every causal claim is registered against the stricter covariance-matched null (Appendix A). The private split is a fresh sealed epoch of 1,024 series, opened once. A new external leg re-tests the registered transfer claims on 800 raw windows from 12 GIFT-Eval datasets (Aksu et al., 2024). These windows are cut unchanged, with no generator, and share no sample with either split; the leg is reported apart from the confirmation verdict.

Of 63 replicable claims, 41 confirm and 3 cannot be tested (Table 5). The pattern of Section 7.7 repeats:

- All 20 transfer claims confirm on the private split, and all 20 again on the raw real windows (AUC 0.757–0.991).
- 11 of 24 testable single-feature causal claims confirm (v2: 10 of 31).

Registered claim type	v2 (971 private series)	V3 (1,024 private series)
Accuracy (Chronos-2 better on `mixture`)	1/1	1/1
Peak-CKA pair	1/1	1/1
Perturbation-profile agreement	6/9	4/9
Concept transfer (shared inputs)	20/20	20/20
same claims on 800 raw real windows	–	20/20
Single-feature causal effect	10/31	11/24
Atlas structure	2/2	2/2
Failure prediction from internals	2/2	2/3
Cross-model causal (atlas, agreement)	0/3, 0/6	none registrable
Confirmed / replicable (not testable)	43/82 (6)	41/63 (3)

Table 5: Held-out confirmation on two independent corpora. Each cell is confirmed / tested; each claim type is one Holm family. The external raw-real leg (800 GIFT-Eval windows) is not counted in the totals. Sources: `confirm/confirmation.json` of `panel7_v2` and `panel7_v3` (CA-11, CA-12).

- Both structural claims confirm: no private atlas concept spans more than 2 of 7 models, and 0.919 of 861 prominent features are causally null.
- Failure prediction confirms again for Timer and Time-MoE; Chronos-T5-Base, newly registered, does not.
- Chronos-2 is again better than TimesFM on `mixture` (0.303 [0.176, 0.445], $p_{\text{Holm}} = 0.0002$).
- The peak-CKA pair replicates (private 0.449 [0.435, 0.464], dev 0.444).

Perturbation agreement replicates for 4 of 9 corruptions under the strict rule, and 8 of 9 keep their dev sign. Level shift flips sign again (dev -0.164, private 0.252), so it is the one depth agreement that is not stable across corpora. No cross-model causal claim was registrable: all four atlas concepts sat on targets that failed the per-target significance gate set before the data ($q \geq 0.243$). Cross-model results barely depend on the alignment window: re-pooled at 64 and 96 steps, the transfer rate moves from 0.827 to 0.801 and 0.799, and 92% of test verdicts are unchanged (SH-23).

7.9 What limits the causal evidence

Fewer than half of the single-feature claims replicate on either corpus, and under the stricter null the number of V3 features that clear at least one channel is below what independent channels would give by chance (91 against 111.2). V3 tested three ways of strengthening this evidence. Each was judged by a criterion set in advance, and none raised per-feature specificity.

- **More training data for the dictionaries.** Three times the rows raised the median held-out fidelity from 0.749 to 0.82. Across targets, however, fidelity is if anything negatively correlated with the share of tested features that clear the null ($\rho = -0.326$, $p = 0.022$, $n = 49$; MN-34).

- **A powered per-feature test.** Using 32 series and 64 null draws per feature, with multiplicity control, fails the planted-concept gate that the 8-series rule passes (sensitivity 0.6 against 0.911; MN-36). The null is common-mode across series, so extra rows barely tighten it.
- **Choosing candidates by gradient attribution.** The ranking (|activation $\times$ gradient of each horizon step|) first passed its own planted gate on unseen seeds (recall 0.831). It nominated 620 more features and found more causal ones (78 to 112 on 38 targets), but at no higher rate than random candidates (0.085 against 0.110, $p = 0.335$). Against the isotropic null it does beat random candidates (0.299 against 0.192, $p = 0.002$; MN-37).

The binding limit is therefore the size of a feature's effect relative to on-manifold directions, not the amount of data, dictionary quality or which features are tested. Since fidelity does not predict causal clarity, the dictionary admission gate now uses fidelity only as a 0.5 sanity floor, with forecast preservation measured on 128 series; under that gate 35 of 50 V3 dictionaries are admitted instead of 8, without retraining (MN-34).

A separate question is how much of the forecast change the dictionary can express is carried by a few atoms. On 31 targets, zeroing the five attribution-ranked atoms of a layer produces a median 0.74 share of the combined feature-plus-dictionary-error change. On 30 of the 31 targets this exceeds activation-matched random five-atom sets (descriptive; D-42C). The forecast change a dictionary can express is concentrated in a few atoms, but those atoms mostly do not act more specifically than a typical on-manifold direction of the same size.

RQ	Result (example run unless an ID is given)	Evidence class	Status	ID
1	194 causal features in 55 targets; 442 cells = 2.47× empirical chance; Timer 1.02×	causal within-model	exploratory	MP-07, PM-17
1	9 families cover 99% of features but do not beat a shuffle null ($p = 0.68$)	descriptive	exploratory, 2 corpora	CA-10
2	Level raise/lower and seasonality-damping causal features in all 7 models	causal within-model, descriptive	exploratory	CA-10
2	Peak CKA 0.426 (private 0.411) vs null 0.038	geometric	held-out confirmed (2 corpora)	SH-01
2	7 similarity metrics agree only weakly ($W = 0.25$)	mixed	exploratory	SH-05/06
2	Untrained pair more similar than related trained pair	geometric	confirmed negative	MN-01
2	Concept transfer on inputs: 20/20; V3 20/20, raw real windows 20/20	correlational	held-out confirmed (5 draws)	SH-18
2	No concept spans the panel (at most 3 of 7 models)	descriptive	held-out confirmed (2 corpora)	SH-16
2	Causal atlas organized by model ($p_{\text{below}} = 0.025$)	descriptive	exploratory	SH-22
2	Multi-model atlas concepts: 0/3	causal, compared	held-out not confirmed	CA-11
2	Same effect on same inputs: 7/216 scorable, all Chronos-2/Bolt (test specific: 0/9 decoys)	causal on shared inputs	exploratory	SH-22, MN-30
3	Single-feature causal effects: 10/31; V3 11/24	causal within-model	held-out partly confirmed	CA-11/12
3	Prominence ≠ importance: 84% of prominent features causally null (927 held-out)	causal within-model	held-out confirmed	MN-14
3	Feature effect 3.5× below SAE reconstruction cost (earlier run)	causal within-model	exploratory	MN-15
3	Attention mass uncorrelated with head causality	causal vs. descriptive	exploratory	CA-04
3	Internals predict failure for the two weakest models only	predictive	held-out confirmed (2 corpora)	MN-28
3	More data, a powered test or attribution candidates do not raise specificity	causal within-model	negative	MN-34/36/37
4	Depth-profile agreement, 6/9 corruptions; V3 4/9, level shift flips on both	causal within-model	held-out partly confirmed	CA-01
4	Captured compute 14%–99.8% qualifies depth claims	descriptive	confirmed	DE-05
L0	Chronos-2 more accurate than TimesFM on `mixture` (0.209; V3 0.303)	behavioral	held-out confirmed (2 corpora)	PM-05

Table 6: Headline results with evidence class and status. Each ID is an entry in `FINDINGS.md` with exact numbers, reproduction command and artifact path.

8 Using and Extending TSFM-Lens

A typical session installs the two packages, builds a benchmark and runs one configuration:

```
pip install -e . && pip install -e tsfm_lens
cd tsfm_benchmark && PYTHONPATH=. python3 example_runs/run_full.py \
       --config configs/example.yaml --out ./benchmark_out --references monash
cd ../tsfm_lens && python run.py --config configs/examples/medium_2model.yaml
```

The CLI adds a preflight check (`--doctor`), partial reruns (`--stages`, `--force`), checks for new checkpoints or library upgrades (`--check-alignment`, `--check-adapter`) and an environment diff (`--verify-provenance`). A CPU-only mock configuration runs every stage in minutes in continuous integration, and the worked example (`tsfm_lens/docs/worked_example.md`) walks through a report. The seven-model analysis records 21.1 GPU-hours of stage time, mostly SAE training and the concept battery; stages through L3 take about 1.4 hours. Of eight TSFM checkpoints probed, two (Timer, Time-MoE) load through the zero-code path and five with standalone packages need a hand-written adapter (MN-25), for which a template, a conformance check (`--check-adapter`) and a guide with a pitfall checklist (`tsfm_lens/docs/ADDING_A_MODEL.md`) exist. New report sections declare their artifact requirements, new claims must use the typed `Finding` record, and new configuration fields declare which artifacts they affect.

9 Limitations

Models. Seven transformer models with contiguous tokens; disjoint-lag tokens (Lag-Llama) get reduced scope. Timer's features do not exceed chance, and with seven models, contrasts are confounded with architecture, size and training data. The analysis is univariate: Chronos-2's cross-series group attention is not measured (PM-13), and Chronos-T5's decoder and Sundial's flow head are not captured.

Ground truth, external validity and confirmation. Ground truth is exact only for the synthetic tier and encodes our generators' assumptions; 12 of 33 concept parts are dominated by one real-derived generator, and pretraining overlap cannot be verified. Real data are part of the analysis, not absent from it: 410 of the 965 dev series are real-derived (Monash weather, ETT, and Chronos electricity and traffic) and the dictionaries are trained with 4,000 additional real Monash windows. Raw real benchmarks are nonetheless a weaker bed for the causal questions: real rows give TimesFM representations of lower effective dimension than the synthetic corpus (1.76 against 2.66) and more dead dictionary atoms (MN-20); weakly predictable real-derived series receive flat forecasts in 67–100% of cases, where an ablation has nothing to move (BM-06); real series carry no generative ground truth; and public archives may be in the models' pretraining data. Raw real data enter as an external check: the V3 transfer claims confirm on 800 raw GIFT-Eval windows (Section 7.8), but no concept was discovered on raw real data, and a full discovery run on a large raw benchmark remains to be done. Held-out confirmation covers accuracy, geometry, perturbation agreement, transfer, single-feature causal effects, atlas structure and failure prediction, on two corpora, each with one private split opened once (a second look needs a fresh epoch). The 20 confirmed v2 transfers come from two source concepts; the V3 transfers come from three source models (Sundial 14, Chronos-2 5, Time-MoE 1).

Effect sharing on the same series. The cross-model causal claims are the ones that did not confirm, and the known-answer test explains why: the shared-input rung is specific but detects a planted shared concept across architectures in at most 0.4 of seeds, and more shared series do not help (MN-30). We therefore built the redesign this diagnosis suggested (ablating a concept's member subspace on each side, an equal-rank size-matched null, and interval estimates against matched floors) and fixed its pass criterion in advance: sensitivity ≥ 0.8 on both planted shared concepts, false agreement ≤ 0.05, on held-out seeds. It also fails (sensitivity 0.5 and 0.375, with 0 of 14 false agreements), and the reason is a ceiling rather than a design detail: even the planted distributed subspace itself does not move the forecast more than an equal-sized random removal (MN-32). Per-series agreement of causal effects across architectures is therefore not measurable by ablation at the effect sizes these models have, and TSFM-Lens compares effects across models at the level of concept profiles instead (Section 7.3).

SAE dependence, usability and completeness. Concept results depend on dictionary size, sparsity, revival and seed; the system measures this dependence but cannot remove it, and feature death tracks layer geometry (alive atoms scale with effective dimension with log-log exponent 1.02, $\rho = 0.58$ over 45 cells; MP-05). Readability for newcomers is a design intention; no user study measured comprehension. Controls catch only failures for which a known-answer case exists.

10 Conclusion

TSFM interpretability is harder than it looks: tokens mean different things in different models, there is no annotated ground truth, and causal effects are small enough that measurement artifacts dominate unless controlled. TSFM-Lens addresses this with a controlled benchmark, measured alignment, null- and reach-gated causal concepts and a register-then-confirm protocol, behind one command and a report designed to be understandable by newcomers. On seven TSFMs, causal features form a compact vocabulary centered on forecast level and seasonality; different architectures select the same inputs for a concept, replicably across four sealed draws, while organizing their causal machinery by model; causal importance sits in a minority of features that activation prominence does not reveal; and depth signatures replicate held-out. Each of these conclusions is stated with its evidence class, and each held-out claim was registered before the private data were opened.

Availability. Code (MIT), configurations, the example run and the findings ledger are at `https://github.com/DanK10097110/TSFM-Interp`; the `dev` branch holds the full research log and study drivers. Version 1.0.0 is archived at doi:10.5281/zenodo.23267463, and both packages install from PyPI (`pip install tsfm-lens tsfm-benchmark`).

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the experiments and helped draft and edit the manuscript, under the authors' direction. The research questions, ideas and general design choices are the authors', who reviewed the implementation details Claude proposed, checked every reported result and take full responsibility for the content.

Appendix A. Questions a Reader May Ask

Each answer gives the evidence, its source, and what remains unmeasured. Numbers are from the seven-model run unless an identifier says otherwise.

Is the benchmark trustworthy? Every sample passed a banded-DTW leakage gate (threshold 0.35) against Monash reference series: 68 of 2,000 candidates were rejected, and the closest accepted sample sits at distance 0.352. The dev split and the private epoch used for confirmation are exchangeable: energy-distance $p = 0.951$, 20 of 22 catch22 features equivalent by TOST (the other two inconclusive at this n), tier and family composition indistinguishable ($p = 0.94$ and $p = 0.9999$), and 6 near-duplicate pairs among 937,015 cross-split pairs. Diversity is measured in catch22 space: the dev split has effective dimension 5.37 of 22 features (private 5.57) and a near-collision fraction of 0.052. Each sample is labeled with its tier, so every result can be split by tier. The V3 corpus passes the same checks: its dev split (2,876 series) and private epoch (2,868 sealed series, of which `confirm` draws 1,024) give energy-distance $p = 0.995$, all 22 catch22 features equivalent by TOST, tier composition $p = 0.97$, and 9 near-duplicate pairs among 8,248,368 cross-split pairs. This check is model-free and was computed after the confirmation look. The realism report compares raw moments and is not informative across scales; its only usable statistic, spectral centroid, differs from the reference by 0.022.

Why a synthetic benchmark rather than raw real data? Four reasons, each measured. Ground truth: only generated series record their components, so only they can validate a probe (P1). Representation: on real Monash rows TimesFM's representations have lower effective dimension than on the synthetic corpus (1.76 against 2.66) and more dead dictionary atoms (MN-20). Measurability: weakly predictable real-derived series receive flat forecasts in 67–100% of cases (BM-06); across ablation panels, noise-like contexts are flat 82.3% of the time against 8.5% for smooth ones, and a flat forecast gives an ablation nothing to move. Contamination: public archives may be in the models' pretraining data. Real data are still used, each for a stated purpose: real-derived series (410 of 965) for realism and per-tier checks, about 4,000 raw Monash windows per target only as extra dictionary-training rows, and the Monash archive only as the leakage reference. The V3 corpus labels each series with one of these roles and adds a fifth: 800 raw GIFT-Eval windows held out from every dev stage, used only to re-test the registered transfer claims (20 of 20 confirm).

Is there enough data? It depends on the claim, and each claim's power is stated. Not power-limited: the 20 transfer claims all reach the Holm floor (AUC 0.919–0.970 against null 95th percentiles of 0.70–0.75); the accuracy claim's minimum detectable ΔMASE is 0.133 against an observed 0.209; the CKA interval is $[0.401, 0.427]$ against a null of 0.038. Power-limited: single-feature causal claims (each feature is tested on its 8 strongest-firing series, and 19 of the 21 that did not confirm never reached the dev p floor), per-target concepts (178 of 672 candidates never fire on any series, so more rows do not help; SH-20),

and the shared-input agreement rung, which more shared series do not rescue (MN-30). More data was then tested directly. A three times larger corpus (V3, 2,876 dev series) reproduces the held-out pattern (41 of 63 claims) but does not make features more causal. Neither does a per-feature test with four times the series and null draws, which is less sensitive on the planted forecaster (Section 7.9; MN-34, MN-36).

Do the models forecast sensibly on this corpus? Five of seven beat the seasonal naive forecast on 77–85% of series (Chronos-2 0.845, TimesFM 0.823, Sundial 0.798, Chronos-Bolt 0.784, Chronos-T5-Base 0.765); Timer does on 0.695 and Time-MoE on 0.378 (Table 4). The models are therefore analyzed while doing the task they were trained for, except Time-MoE, whose results the paper reads with its accuracy in mind.

Why define concepts by causal effect rather than by activation or correlation? Because correlation misleads here, measurably. 83.5% of the most prominent features have no causal effect on held-out data (MN-14); attention mass on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated (CA-04); two sparse features can correlate at $\rho = 0.80$ with no shared top series (MN-19); features aligned with a ground-truth field do not mediate that field's effect (MN-17); and on a planted forecaster a decoy concept correlates with forecast error in five of five seeds while its causal effect is null (MN-31). An activation-based concept can be real and still irrelevant to the forecast; the causal definition asks the question that matters for forecasting.

Is zeroing a feature a fair intervention? For a TopK dictionary, zero is a feature's resting value: at most 32 of 192–6,144 atoms are nonzero on any token, so the ablated state lies inside the dictionary's normal range, unlike zeroing a dense neuron. The comparison is between two passes through the same reconstruction, so the dictionary's error is in both arms. The null removes exactly the feature's per-token amount along another direction, on the same series, with the same seed for sampled models, after a reach probe has shown that patching this layer can move the forecast. Two alternatives were not run: mean ablation, and ablating against the clean activation; the latter differs only through the model's nonlinear response to the reconstruction error.

How do you know a causal feature is not chance? Four checks. (i) Calibration on a known answer: on the planted forecaster's control layer, which has no planted effect, the per-cell false-positive rate is at most 0.003 while planted effects are found with sensitivity 0.70–1.00 (MN-29). (ii) Counting against an empirical chance rate, estimated per cell by scoring each null draw against the others (median 0.018): 442 feature-channel cells clear against 178.75 expected (2.47 times chance), and 22 of 55 targets are individually significant (BH). At the level of whole features the excess is smaller: 194 of 1,022 scorable candidates clear at least one channel against at most 144.95 expected if channels were independent. (iii) Held-out replication: 10 registered single-feature claims confirm on new series at the attainable p floor after adaptive redraws to 1,000 null draws. (iv) Null strictness: a stricter null that draws the random direction from the layer's own activation covariance also passes the planted test (sensitivity 0.916), but on the earlier four-model run it leaves only 1.13 times chance. Causal effects therefore exceed random directions clearly and typical on-manifold directions only slightly. The V3 run registered every causal claim against the stricter null. There, feature-level clears fall below the independent-channel expectation (91 against 111.2),

yet 11 of 24 registered claims replicate on new series (CA-12). The stricter null leaves few causal features, and about half of the registered ones replicate.

How large are the effects? Small relative to the forecast and to the dictionary's own error. On the earlier four-model run the median single-feature effect was 0.0395 context standard deviations, 3.5 times below the change from substituting the reconstruction (MN-15); among the cells that clear in the seven-model run, the median effect is 1.24 times the null's 95th percentile. The paper therefore makes no claim that single features explain forecasts; it claims that specific, replicable directions exist and characterizes what they do.

How robust are the sparse dictionaries? Each target has three replicate dictionaries, and 9 of 11 atlas concepts are stable across them. Held-out fraction of variance explained has medians of 0.647 (TimesFM) to 0.878 (Chronos-T5-Base) across models, and only 5 of 56 targets pass the strict admission gate (held-out fidelity ≥ 0.85, forecast change under substitution ≤ 0.25 MASE, dead-rate bound). With three times the training rows (V3), the median held-out fidelity rises to 0.82, but fidelity does not predict causal clarity ($\rho = -0.326$ across 49 targets; MN-34). The ablation keeps the dictionary's error in both arms, so a measured effect is a real intervention whatever the fidelity. Fidelity is therefore now only a 0.5 sanity floor, and forecast preservation is measured on 128 series; under this gate 35 of 50 V3 dictionaries are admitted. Joint crosscoders were tested as the alternative and lost to independent dictionaries matched afterwards (0.2787 against 0.3204, $p = 0.002$; SH-10), which is why TSFM-Lens uses one TopK dictionary per layer.

How are token time frames aligned across models? In three steps, all measured rather than declared. (1) Each token's span is measured by injecting an impulse at every time step and recording which token responds most, gated on peak-to-pedestal contrast (≥ 4) and contiguity (≥ 0.95); measured spans equal the hand-written adapters' on six checkpoints (IoU 1.0) and recover Timer's 96-step tokens (MP-01). (2) Tokens are overlap-pooled onto common 32-step windows (Equation 1): Chronos-T5's one-step tokens are averaged 32 to a window, TimesFM's 32-step tokens map one to one, and each Timer token covers three windows, which then share one value (the alignment gate's ceiling for Timer is therefore one third). The context of 480 steps is a multiple of 96 so that Timer's tokens tile it exactly (at 512 its span discovery refuses). (3) An alignment gate injects an impulse into each window and checks per layer that the matching window moves most; all models pass at their shallowest layer, and the gate's per-layer record shows that deep layers of Sundial, Time-MoE and Timer lose time localization (diagonal hit 0.07–0.27 against chance 0.067), which the report renders and which qualifies window-level claims there. Values are normalized by each model's own input scaling, verified by a scale-equivariance test (residuals 2.7×10^{-7} to 0.037 context standard deviations); effects are compared across models in units of each feature's own null 95th percentile; depth is the fraction of the whole stack. Re-pooled at 64 and 96 steps (V3), the transfer rate moves from 0.827 to 0.801 and 0.799, with 92% of verdicts unchanged, and peak CKA by a median of 0.013 and 0.018. The peak *layer pair* is the same at all three windows for only 9 of 21 model pairs, so peak locations are window-dependent (SH-23).

Do sampled models make the causal comparison noisy? Every compared pair of calls to Sundial or Chronos-T5 uses the same seed. Unseeded, back-to-back Sundial calls

differ by up to 0.50, and a null seeded differently from its baseline once made Sundial look causally inert (PM-09); repeat-run noise floors are reported per model (Sundial 0.14, Chronos-T5-Base 0.24 mean |ΔMASE|, the deterministic models 0).

Could the private split have influenced the claims? The registry (128 hypotheses, sha256 recorded) was frozen before the private epoch was generated; the epoch shares no sample hash with any of seven earlier splits (one candidate epoch was discarded unopened for overlapping an old split); `confirm` ran once, and a crash before any statistic was written is disclosed in Appendix D. Registered claims are cuts from larger dev pools (transfer 20 of 1,124 candidates, causal 32 of 113, agreement 11 of 25, atlas 4 of the 9 stable concepts), which is legitimate because the private split referees them, and is why only registered claims are called confirmed. The V3 registry (111 hypotheses, sha256 `67b91269...`) was frozen in the same way before its private epoch was minted; its external windows share no sample hash with the dev or private split.

How are multiple comparisons handled? Dev results are exploratory and labeled so. The ledger groups 4,238 tests into 123 named families with Holm or BH and flags families that cannot reach significance; the battery's 8,705 cells are judged by their count against the empirical chance rate. Held-out claims use Holm per registered claim type.

Why CKA, given its known problems? It is used only against a shuffled-series null at the same layer pair with a series-level bootstrap, reported as one of seven pair metrics that are never pooled, and never read as lineage, since an untrained pair can score higher than a trained one (MN-01). The reported peak is a maximum over layer pairs; its held-out replication (0.411) is measured at the pair selected on dev.

Were settings chosen after seeing results? Battery settings ($k = 8$ rows, 16 null draws) were fixed on the four-model run before the seven-model concept stage ran, after larger settings did not recover power (SH-20); at the atlas threshold of 0.9 the real concept count was about twice its shuffle-null mean on that earlier run (19 against 9.62); family silhouettes are flat (0.25–0.32) across cuts from 6 to 16, which is why families are reported as a reading vocabulary.

Does this generalize to other models? The pipeline is model-agnostic behind the adapter interface, and two of the seven models load through the zero-code path. Of eight checkpoints probed, five crashed in their configuration loader and need hand-written adapters (MN-25); models whose tokens read disjoint lags (Lag-Llama) receive behavioral analysis only.

Appendix B. Findings Ledger Index

Each identifier is an entry in `FINDINGS.md` giving exact numbers, models, evidence class, status, a reproduction pointer (configuration, command, artifact path), evidence class and status, and an interestingness score (1–5); negative results are included on equal terms. The table abridges each claim; evidence class and status are in the ledger. Prefixes: SH shared vs. unique; CA causal agreement; DE depth; PM per-model character; MP method positives; MN method negatives; BM benchmark trust.

Table 7: Ledger entries cited in this paper (score = interestingness, 1–5). *Section (§) or table (T) of this paper where the result is reported.

ID	Claim (abridged)	In paper*	Score
SH-01	Peak CKA 0.38–0.47 above shuffled-series nulls (0.02–0.16) across five architectures	§7, T6	5
SH-05	CKA and clustering AMI rank model pairs differently on three panels	T6	5
SH-06	Pair similarity splits into a geometric camp and a behavioral/causal camp; 7 metrics never pooled	T6	5
SH-10	Joint crosscoder loses to independent SAEs matched post hoc (0.2787 vs 0.3204, $p = 0.002$)	App. A	4
SH-16	No atlas concept spans the panel (at most 3 of 7 models; held out)	§7, T6	5
SH-18	Registered concept-transfer claims confirmed on private data: 19/20, 20/20 frozen, 20/20 on v2, 20/20 on the 7-model epoch 3, 20/20 on V3 and on 800 raw real windows	§7, T6	5
SH-19	Unit-level correspondence fails rotation controls (0.148 → 0.331); participation ratio as low as 1.54	§7	5
SH-20	Per-target concepts mostly non-modular: 50 of 56 targets at 7 models	§5, §7	4
SH-23	Transfer and peak CKA barely change at windows 64 and 96; peak layer pair stable for 9 of 21 pairs	§7.8, App. A	3
SH-22	Atlas segregated by model; all 7 "same causal effect" verdicts are Chronos-2/Bolt; rung insensitive	§7, T6	5
CA-01	Perturbation-fingerprint agreement is pair-specific; TimesFM/Chronos-2 replicates on private data	T6	4
CA-03	Seasonality circuit: 1 head (TimesFM) vs. 5 non-additive heads (Chronos-T5-Base)	§7	4
CA-04	Attention mass at seasonal lags uncorrelated with causal importance ($\rho = 0.026$, -0.118)	§7, T6	4
CA-07	Median level share of a causal feature's effect 0.535; most features shape-causal beyond a level-removed null	§7	4
CA-10	Causal effect space is a continuum; families tile it but do not beat a shuffle null	§7, T6	4
CA-11	Held-out: 10/31 single-feature causal claims confirm; no cross-model one does (atlas 0/3, agreement 0/6); 43/82 in all	§7, T6	5
CA-12	V3 held-out: 41/63 claims; single-feature causal 11/24; transfer 20/20 private and 20/20 raw real	§7.8, T5	4
D-42C	Five attribution-ranked atoms carry a median 0.74 share of expressible forecast change; beat matched random sets on 30/31 targets	§7.9	3
DE-01	Whole-layer patching hides a depth-intensifying recency effect in the last pre-horizon window	§7	4
DE-03	Chronos-2 skip lens was a silent no-op (identical MASE at 12 layers); corrected reach 0.737 → 0.000	§5, §7	5
DE-04	Chronos-T5 encoder ends at depth 0.478 on a whole-stack axis; 50% comparable overlap with TimesFM	§5, §7	5
DE-05	Captured share of forecast FLOPs: 13.9% (Chronos-T5-Base) to 99.8% (Timer) at 7 models (14.35% to 97.5% on four)	§4, §7, T6	5
DE-08	Layer selection depends almost entirely on the selector (0/3 robust layers)	§7	4
PM-05	Chronos-2 most accurate and best observed; winners vary by family; held-out on v2 and on 7-model epoch 3	T6	4
PM-07	Models differ ∼5.4× in reversion to the mean; flat forecasts driven by weakly predictable data	§2	4
PM-09	Sampled models need identical seeds in compared calls; a mis-seeded null made Sundial look inert (1/71 → 12/71)	App. A	3
PM-11	Causally load-bearing TimesFM random-walk concept fails seed stability	§7	3
PM-13	Chronos-2 group attention unmeasured by design	§8	3
PM-17	Timer's SAE features do not clear the ablation null above chance (1.017×; 0 of 6 targets)	§7	3
MP-01	Token spans can be measured (IoU 1.0 on 6 checkpoints); zero-code adapter follows	§4	5
MP-02	Alignment gates must be normalized by their attainable ceiling	§4	3
MP-03	AuxK revival recovers dead SAE atoms 10–13× on real activations	§5	4
MP-04	Single-seed SAE gold ranking flips a bake-off verdict; 3 seeds fix it	§5	5
MP-05	SAE feature death set by layer geometry (exponent 1.02), not recipe	§8	5
MP-06	Ground-truth alignment of SAE features is real only over a permutation null	§7	3
MP-07	Ablation battery discriminates: 2.47× empirical chance at 7 models (Timer 1.02×, Bolt 5.70×)	§7	4
MP-09	Dev and private splits exchangeable (energy $p = 0.51$; TOST 21/22)	§4	3
MN-01	Similarity cannot detect lineage: untrained pair 0.878 > trained related pair 0.734	§7, T6	5
MN-07	Causal fingerprints are nearly one-dimensional (PR 2.09)	§7	4
MN-14	0.835 of 927 prominent SAE features are causally null on private data (dev 0.810; 64.4% under the legacy null)	§7, T6	4
MN-15	SAE reconstruction moves the forecast ∼3.5× more than removing a feature	§7, T6	4
MN-17	Generator-counterfactual concept mediation: 0/98 and 0/151	§7, T6	5
MN-18	Absence of agreement is not disagreement: 109/288 → 9	§5	4
MN-19	"Same inputs" must be top-k overlap, not whole-series correlation	§5	4
MN-20	Real Monash rows: lower effective dimension (1.76 vs 2.66) and more dead SAE atoms than the synthetic corpus	§9	3
MN-23	MASE denominator floors out on intermittent series	§5	3
MN-25	Zero-code adapter resolves 2 of 8 probed TSFM checkpoints	§8	3
MN-28	Internals beat a free failure-prediction baseline only for the two weakest models; routing does not help	§7.6	4
MN-29	Legacy ablation null fails a planted-concept gate (FPR 0.168–0.248); profile-matched null passes ($\leq$0.003)	§5.5	5
MN-30	Agreement rung is specific (0/9 false "same") but insensitive (0.4 and 0.0 of seeds)	§5.5, §7	5
MN-31	A planted decoy correlates with forecast error in 5/5 seeds but has no causal effect	App. A	3
MN-32	Subspace redesign of the agreement test fails its pre-fixed gate; planted distributed subspace does not clear a size-matched null	§9	4
MN-34	SAE fidelity does not predict causal clarity ($\rho = -0.326$, V3); 3× rows raise fidelity 0.749 → 0.82	§7.9, App. A	3
MN-36	Powered per-feature test (32 series, 64 draws) fails the planted gate (0.6); 8-series p95 rule passes (0.911)	§7.9	4
MN-37	Gradient-attribution candidates: no higher causal rate than random under the covariance null (0.085 vs 0.110)	§7.9	4
BM-05	Counterfactual identity check caught a corruption-replay bug (244/298 → 0/1113)	§4	2
BM-06	42% of the dev corpus is weakly predictable real-derived data, driving the flat-forecast share	§9	4

Appendix C. Statistical Details

Paired series bootstrap. With per-series differences $d_s = m_{a,s} - m_{b,s}$ and $B = n_{\text{boot}}$ resampled means $\bar{d}_b^*$, the interval is the percentile interval and $p = \min(1, \max(\frac{1}{B}, 2\min\{\frac{1}{B}\#(\bar{d}_b^* \le 0), \frac{1}{B}\#(\bar{d}_b^* \ge 0)\}))$ (`analysis/stats.py`; floor per Phipson and Smyth, 2010). Window-level statistics (CKA, fingerprints) resample series with all their windows (Cameron et al., 2008).

Multiplicity. Holm: $\tilde{p}_{(i)} = \max_{j \le i} \min(1, (m - j + 1)p_{(j)})$ (Holm, 1979); with $p \ge 1/B$ the smallest attainable value is m/B, which the ledger reports per family, flagging $m/B > \alpha$. BH: $\tilde{q}_{(i)} = \min_{j \ge i} \min(1, \frac{m}{j} p_{(j)})$ (Benjamini and Hochberg, 1995).

Profile-matched ablation null. For feature i with rows R_i and $J = 16$ unit directions u_j, the null removes at every token t the amount $|z_i(t)|\,\|w_i\|\,u_j$ (the legacy null removed one batch-averaged amount $\bar{a}\,W_d u_j$); channel c clears if $\frac{1}{|R_i|}\sum_{r \in R_i} |\Delta^{(c)}_{i,r}| > q_{0.95}\{|\Delta^{(c)}_{u_j,r}| : j \le J, r \in R_i\}$. A null with no spread is reported as degenerate and not scored.

Transfer. The statistic is $\max_f \mathrm{AUC}(a_f; S)$ over destination features; its null recomputes the same maximum on stratum-matched random k-sets, so the feature search is inside the null. The reverse leg scores the chosen feature's own top-k by the source activation. $p = (1 + \#\{\text{null} \ge \text{obs}\})/(1 + n)$; a registered claim uses $\max(p, p_{\text{rev}})$ (intersection–union), Holm across claims.

Shared-input agreement. On U, statistic (i) is the Spearman concordance of the two sides' signed level effects across series, and (ii) the cosine of their level-removed shape vectors, each channel in its own null units. Each side's floor ablates random feature sets matched to its activation on U and re-scores against the other side's measured effect; agreement exceeds both floors' $q_{0.95}$, "acts differently" falls below both $q_{0.05}$.

Minimum detectable effect. Before the private result is read, recentred per-series differences are shifted by a candidate effect and re-tested with the same bootstrap; bisection finds the smallest shift detected with power 0.8 (`analysis/power.py`).

Appendix D. Reproducibility

`examples/panel7_v2` holds the configuration, the report and every stage's JSON/parquet artifacts plus the confirmation (activations and SAE weights omitted for size); `paper/figures/make_figures.py --run DIR` regenerates all data figures. From `tsfm_lens/`, `python run.py --config configs/panel7_v2.yaml` reproduces the run (add `--stages report` to re-render, and `--verify-provenance runs/panel7_v2_dev` to diff environments). The private split (v2 epoch 3, 971 series) was opened once (2026-10-02), so re-confirming requires a fresh epoch. Two events are disclosed. A first epoch minted for this look (v2 epoch 1) was rejected before any model saw it, because 716 of its 961 series equalled series of a split an earlier confirmation had consumed (epoch seeds depend only on the seed and epoch number). And the first confirm attempt crashed after every private battery finished, while assembling the claim ledger (a fixed family map lacked the failure-prediction family), before any statistic was written: `confirmation.json` did not exist, no private number was read, the registry hash and every frozen specification

were unchanged, and the restart differed only by the crash fix. It is one test of the registry run twice for compute reasons; the artifact records `repeated_look: false`. The V3 run (`tsfm_lens/configs/panel7_v3.yaml`; dev 2,876 series, private epoch 1 of 1,024 series opened once on 2026-10-10, external 800 GIFT-Eval windows) records its confirmation in `runs/panel7_v3_dev/confirm/confirmation.json` (`repeated_look: false`); the step that re-ranks candidates by attribution is the separate, exploratory `configs/panel7_v3_sec42.yaml`, built on the same activations and dictionaries. In the v2 confirmation, 19 of the 21 non-confirmed single-feature claims carry a stored reading of "untestable" in `confirmation.json`; they were tested and not detected, the report recomputes the reading at render time, and no verdict is affected. Environment, as recorded in the run's `run_manifest.json`: Python 3.12.13, numpy 2.1.0, PyTorch 2.12.0 (CUDA 12.9 build), transformers 4.57.6, timesfm 2.0.2, chronos-forecasting 2.3.1, zarr 2.18.7 (zarr 3 stores read back empty under the v2 API), pandas 2.3.3, scikit-learn 1.7.2; load-bearing pins and their reasons are in `DEPENDENCIES.md`. Tests: 2,520 (`tsfm_lens`) and 125 (`tsfm_benchmark`), run in continuous integration. Recorded stage wall-clock time on one RTX A5000: 75,900 s in total, of which `sae` 14,614 s and `concepts` 55,608 s (shared-input agreement 45,189 s); every other stage under 33 minutes.

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